Econ 602 Notes

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Last Updated: 2024-02-02

Simultaneous-move Games of Complete Information

Can model these as strategic-form games

Def "strategic-form game"

A strategic-form game is a $G = (S_i, u_i = 1, ..., I)$ such that

1. $\{1, ..., I\}$ is a set of players
2. S_i is a set of strategies for player i
3. $u_i : \Pi_{j=1}^{I} S_j \to \mathbb{R}$ is a payoff function for player i. (u_i maps from strategy profiles to real (integrable) numbers.)

Example 1: introduce NE

Def "Nash Equilibrium (PSNE)"

Call $(s_1^*, ..., s_I^*)$ a Nash equilibrium if, for each $i = 1, ..., I$

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}), \forall s_i \in S_i.$$

"deviating to some s_i is not going to make me any better off.

Def "Best Response"

Say $s_i \in S_i$ is a best response for i under s_{-i} if

$$u_i(s_i, s_{-i}) \geq u_i(s_i', s_{-i}) \quad \forall s_i' \in S_i.$$

Def "Best Response Correspondence"

$BR_i : S_{-i} \rightrightarrows S_i$ such that

$$BR_i(s_{-i}) = \{s_i \in S_i : s_i \text{ is a best response for } i \text{ under } s_{-i}\}$$

Remark 1 A strategy profile is a Nash Equilibrium if and only if, for each $i = 1, ..., I$

$$s_i^* \in BR_i(s_{-i}).$$

Example 2 Maximize revenue to find BR correspondence. Determine PSNE from correspondence.

Example 3 "Increasing payoffs can hurt"

Difference between Game Theory and Decision Theory: In DT, increasing payoffs cannot hurt. In GT, there are scenarios under which increasing payoffs can hurt.

Example 4 "incentive to hide what you are playing"

Def "set of mixed strategies of i"

FINISH

Remark 2 "infinite strategies have no atoms..." If S_i is finite, then $\sigma_i(E_i) = \sum_{s_i \in E_i} \sigma_i(s_i)$. But, this property need not hold when S_i is infinite.

FINISH

Remark 3

Remark 4

Def "MSNE"

Call $(\sigma_1, ..., \sigma_I)$ a "Mixed Strategy Nash Equilibrium" (MSNE) if, and only if, for each $i = 1, ..., I$

$$u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(\sigma_i, \sigma_{-i}), \forall \sigma_i \in \Delta(S_i).$$

Lemma 1 "MSNE $\iff$ weakly better than all pure strategies"

A strategy profile $(\sigma_1, ..., \sigma_I)$ is a MSNE if, and only if, for each i

$$u_i(\sigma_i^*, \sigma_{-i}^*) \geq u_i(s_i, \sigma_{-i}^*), \forall s_i \in S_i.$$

Lemma 2 "MSNE $\iff$ positive strategy in support is best"

Fix a finite game $G = (S_i, u_i : i = 1, ..., I)$.

Then $(\sigma_1, ..., \sigma_I)$ is a MSNE if, and only if, for each i and for each $s_i \in S_i$, with $\sigma_i^*(s_i) > 0$,

$$u_i(s_i, \sigma_{-i}^*) \geq u_i(r_i, \sigma_{-i}^*), \forall r_i \in S_i.$$

Remark 5 "Lemma 2 doesn't have the right context for the infinite case"

Cor 1 "indifference condition"

Fix a finite game $G = (S_i, u_i : i = 1, ..., I)$.

If $(\sigma_1, ..., \sigma_I)$ is a MSNE and with $\sigma_i^*(s_i), \sigma_i^*(r_i) > 0$, then

$$u_i(s_i, \sigma_{-i}^*) = u_i(r_i, \sigma_{-i}^*).$$

Example 5 "Lemma 2 doesn't work in the infinite case + intuition for how to extend"

Thm "from online notes"

1. If σ_i induces F_i, then F_i is a CDF
2. If F_i is a CDF, then there exists a *unique* σ_i that induces F_i.

Remark 6 "unanswered questions"

Remark 7 "pdf/cdf technical notes" omitted

Thm "Kakutani's fixed point thm"

Let X be nonempty compact convex set. If F is nonempty, convex-valued, and has a closed graph, then F has a fixed point.

Term "Mixed Best Response Correspondence" $\tilde{BR}_i : \Pi_{j \neq i} \Delta(S_j) \rightrightarrows \Delta(S_i) \ldots$ FINISH

Thm 1 "conditions to guarantee existence of PSNE"

Fix a game where, for each $i = 1, ..., I$, if

1. there exists an n_i such that S_i is a nonempty, compact, convex subset of $\mathbb{R}^{n_i}$,
2. $u_i : S \to \mathbb{R}$ is continuous, and
3. for each s_{-i}, holding s_i fixed, $u_i(\cdot, s_{-i}) : S_i \to \mathbb{R}$ is quasiconcave.

   nonempty: from S_i being compact and continuity of utility fn (BR has to be nonempty)

   convex-valued: from quasiconcavity of $u_i(\cdot, s_{-i}) : S_i \to \mathbb{R}$

   closed-graph: from continuity of u_i

Thm 2 "conditions to guarantee existence of MSNE"

For each finite game, there exists a MSNE.

Thm 3 "Glicksberg' thm"

Fix a game, G where, for each i,

1. S_i is compact metric space, and
2. $u_i : S \to \mathbb{R}$ is continuous.

Then G has a MSNE.

Remark 8

Remark 9

Remark 10

Rationalizability and Iterated Dominance

Example 1 " "

Belief of player i: $\mu_i \in \Delta(S_{-i})$